Weak (Proxy) Factors Robust Hansen-Jagannathan Distance For Linear Asset Pricing Models

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Econometrics Group Lunch Seminar

Brief Intro

- HJ distance statistic in linear asset pricing models mainly serves two purposes: (1) model selection; **(2) model specification test.**
- Contribution:
  - We observe that the conventional HJ statistic is unreliable in the presence of weak (proxy) factors, and it can not distinguish between model misspecification and weak identification.
  - We provide alternative specification testing procedures that are robust to the presence of weak (proxy) factors and missing factor structure.
  - Besides, an alternative risk premia estimator.

literature

Linear factor models have gained tremendous popularity in the empirical asset pricing literature, Harvey et al. (2016) list hundreds of papers and factors that attempt to explain the cross-section of expected returns.

Background information: model set up

The traditional HJ statistic is unreliable

Our solutions

We propose two alternative approaches

Empirical application

Conclusion

Literature

This research is closely related to literature on linear asset pricing models, factor models, weak IVs.

- Fama and French (1992),Lettau and Ludvigson (2001), Parker and Julliard (2005), Jagannathan and Wang (2007), Savov (2011), Kroencke (2017), Harvey et al. (2016)
- Bai and Ng (2002), Bai (2003), Giglio and Xiu (2018)
- Kleibergen (2009), Gospodinov, Kan and Robotti (2017), Bryzgalova (2016)
- Fama-MacBeth (1973), Gibbons, Ross and Shanken (GRS) (1989), Kleibergen (2009),Hansen (1982), Beaulieu, Dufour and Khalaf (2013), Khalaf and Schaller (2016), Kleibergen and Zhan (2015), Giglio and Xiu (2018), Anatolyev and Mikusheva (2018)

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Model: beta representation

Linear asset pricing models have two representations: the beta representation and the stochastic discount factor (SDF) representation.

Beta representation:

For the $N \times 1$ vector of asset gross returns r_t, we assume that

$$r_t = c + \beta f_t + u_t,$$

with f_t a $K \times 1$ vector of (possibly) unobserved zero-mean factors and u_t is an $N \times 1$ vector of idiosyncratic components, and with a $K \times 1$ vector of proxy factors g_t

$$f_t = d_g \left(g_t - \mu_g \right) + v_t,$$

where g_t is uncorrelated with u_t, v_t, d_g is of full rank and g_t, v_t, u_t are stationary with finite fourth moments. Furthermore,

$$(*) \quad c = \iota_N \lambda_0 + \beta \lambda_f,$$

where $\lambda_0 \neq 0$ is the zero-beta return, and λ_f is a $K \times 1$ vector of risk premia.

Model: beta representation

Linear asset pricing models have two representations: the beta representation and the stochastic discount factor (SDF) representation.

Beta representation-what is weak (proxy) factors:

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SDF representation (Cochrane (2001)):

$$\begin{aligned} \mathbb{E}\left(m_t(\theta_0) r_t\right) &= \iota_N, \\ m_t(\theta) &= F_t' \theta \end{aligned} \tag{1}$$

with $F_t = (1, f_t')', \theta = (1/\lambda_0, -V_{ff}^{-1}\lambda_f/\lambda_0)$.

HJ distance

The population squared HJ distance δ^2, measuring the deviation from the moment conditions (1), has an explicit expression

$$\delta^2 = \min_{\theta} e(\theta)' Q_r^{-1} e(\theta) = \iota_N' \left(Q_r^{-1} - Q_r^{-1} q \left(q' Q_r^{-1} q \right)^{-1} q' Q_r^{-1} \right) \iota_N,$$

with $e(\theta) = \iota_N - \mathbb{E}\left(m_t(\theta) r_t\right) = \iota_N - q\theta$.

HJ statistic

The HJ statistic, is

$$\widehat{\delta}_g^2 = \min_{\theta_g} \delta_{g,T}^2(\theta_G) = \iota_N' \left(\widehat{Q}_r^{-1} - \widehat{Q}_r^{-1} q_{G,T} \left(q_{G,T}' \widehat{Q}_r^{-1} q_{G,T} \right)^{-1} q_{G,T}' \widehat{Q}_r^{-1} \right) \iota_N,$$

with $\delta_{g,T}^2(\theta_G) = e_{g,T}(\theta_G)' \widehat{Q}_r^{-1} e_{g,T}(\theta_G)$, $e_{g,T}(\theta_G) = \sum_{t=1}^T e_{g,t}(\theta_G)/T = \iota_N - q_{G,T}\theta_G$, $e_{g,t}(\theta_G) = \iota_N - r_t G_t' \theta_G$, $q_{G,T} = \sum_{t=1}^T r_t G_t'/T$, $\widehat{Q}_r = \sum_{t=1}^T r_t r_t'/T$, $G_t = (1, \bar{g}_t')'$.

The estimator resulting from this quadratic optimization problem is

$$\widehat{\theta}_G = \left(q_{G,T}' \widehat{Q}_r^{-1} q_{G,T} \right)^{-1} q_{G,T}' \widehat{Q}_r^{-1} \iota_N.$$

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Simulated critical values

Jagannathan and Wang (1996) propose a model specification test using the **non-pivotal** HJ statistic $T\widehat{\delta}_g^2$. The critical values are drawn from a distribution of weighted sum of independent $\chi^2(1)$ random variables

$$\sum_{i=1}^{N-K_g-1} p_i x_i$$

with x_i being independently $\chi^2(1)$ distributed random variables and p_i being the positive eigenvalues of the matrix

$$S_T^{\frac{1}{2}} \left(\widehat{Q}_r^{-1} - \widehat{Q}_r^{-1} \widehat{B}_g \left(\widehat{B}_g' \widehat{Q}_r^{-1} \widehat{B}_g \right)^{-1} \widehat{B}_g' \widehat{Q}_r^{-1} \right) S_T^{\frac{1}{2}\prime}$$

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Theorem ($\widehat{\theta}_G$ is no longer $\sqrt{T}-$consistent)

Suppose all proxy factors g_t are weak, then

$$\left(\widehat{V}_g(\widehat{\theta}_g - \theta_g) - bias_e - bias_m \right) = O_p(1)$$

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Model selection failures

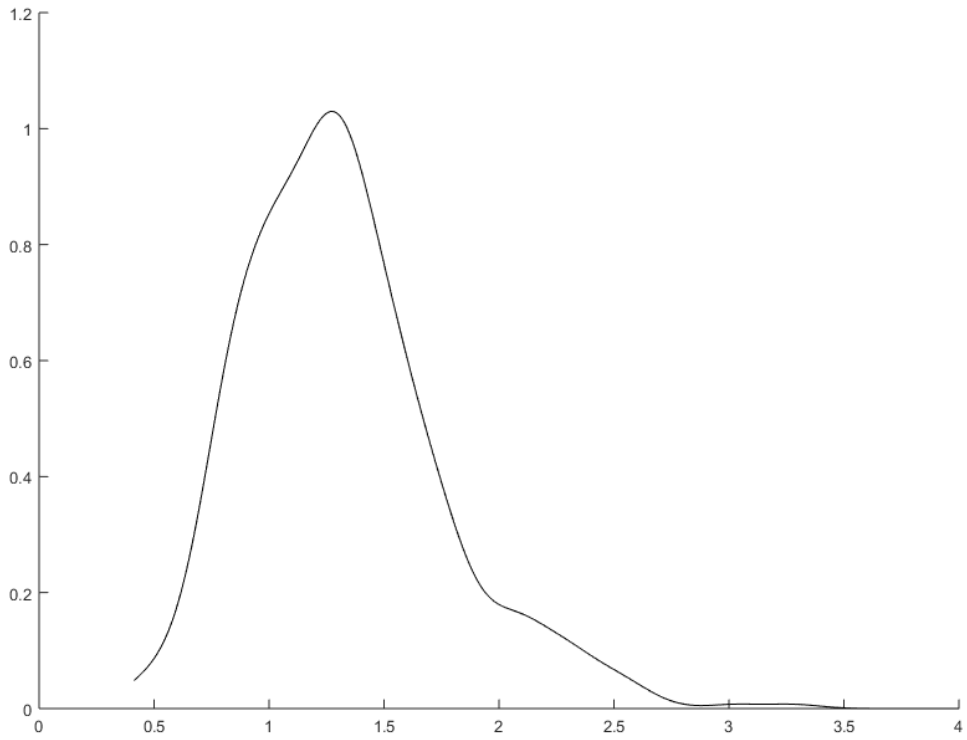
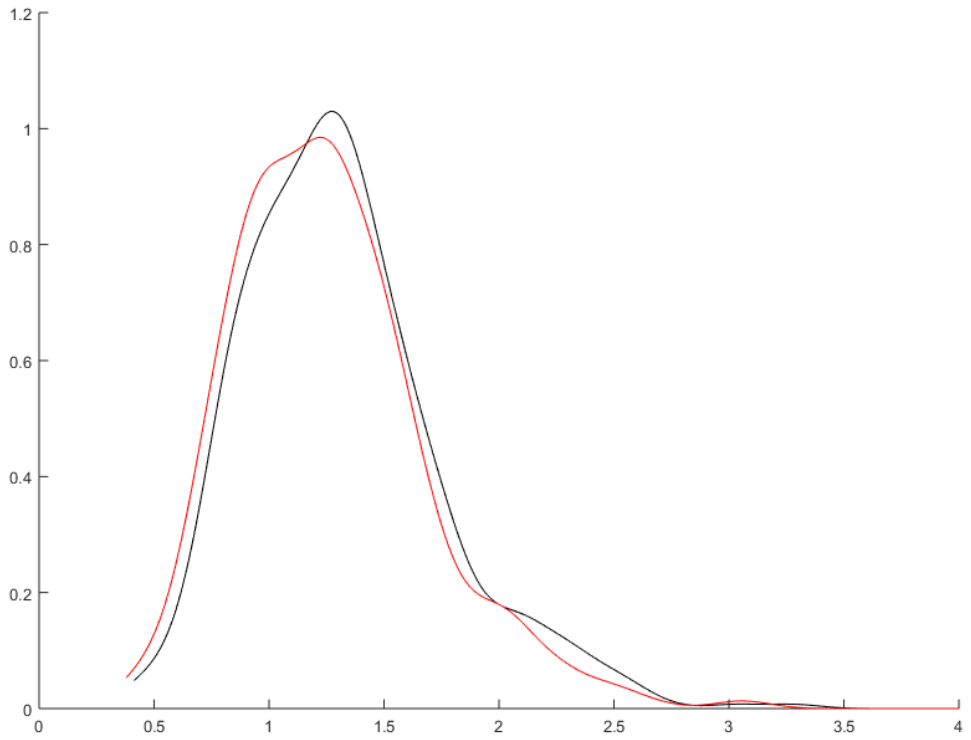


Figure: Density functions of the HJ statistic: (1) black solid: three strong factors; (2) black dashed: two strong factors; (3) black dotted: one strong factors; (4) blue solid: three weak proxy factors; (5) red solid: three strong factors and one useless factor; (6) red dashed: two strong factors and one useless factor; (7) red dotted: one strong factors and one useless factor; (8) red dash-dotted:one strong factors and two useless factors

Model selection failures



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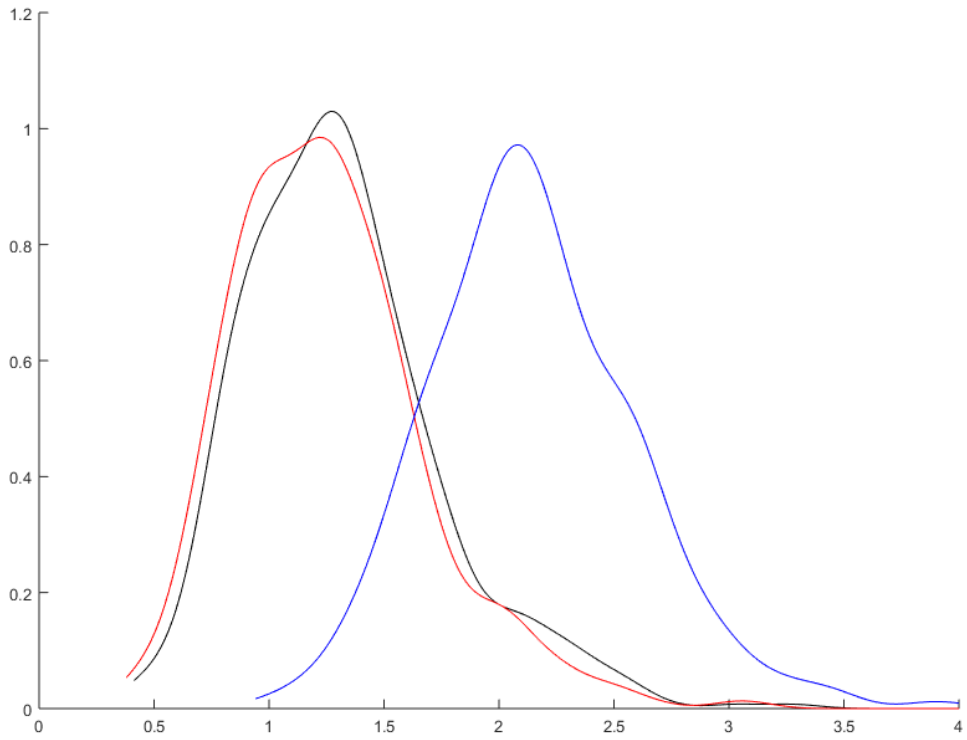

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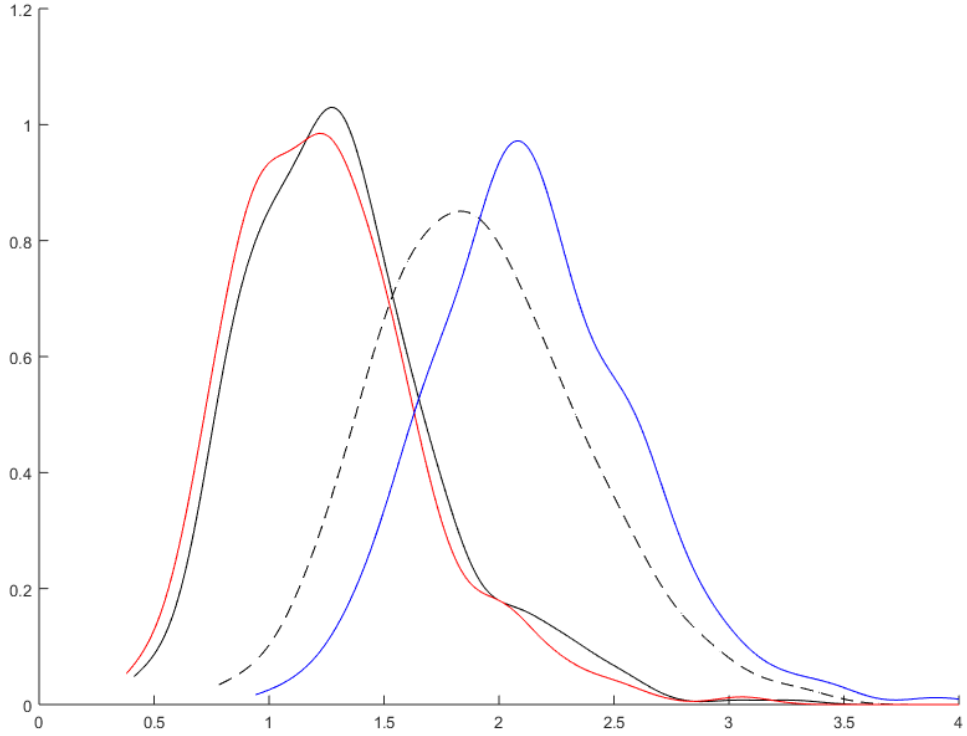

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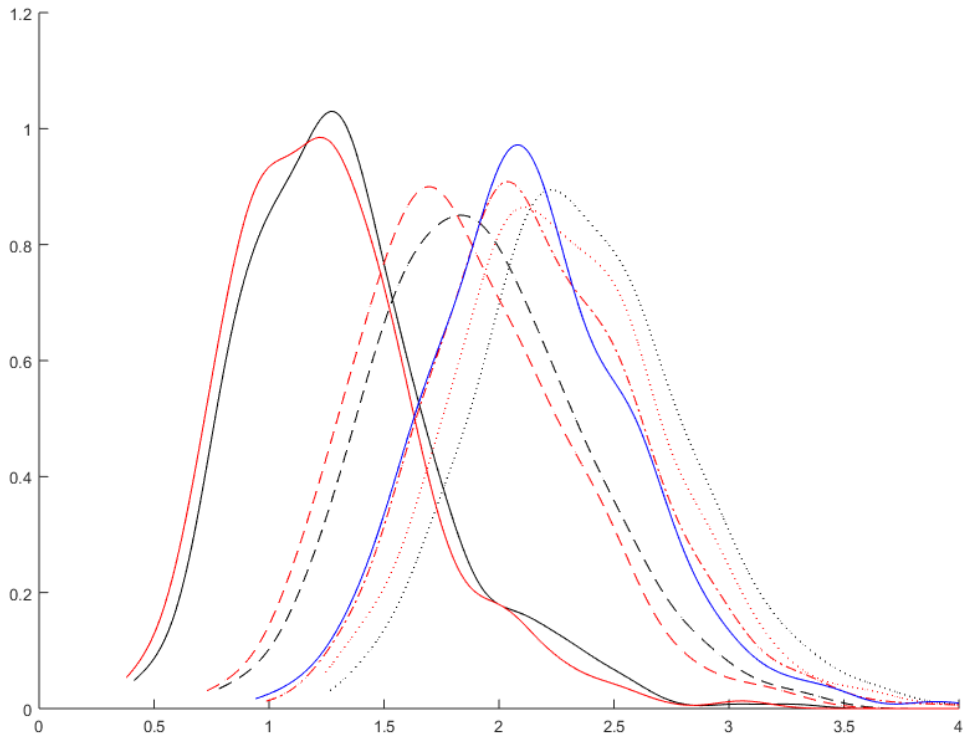

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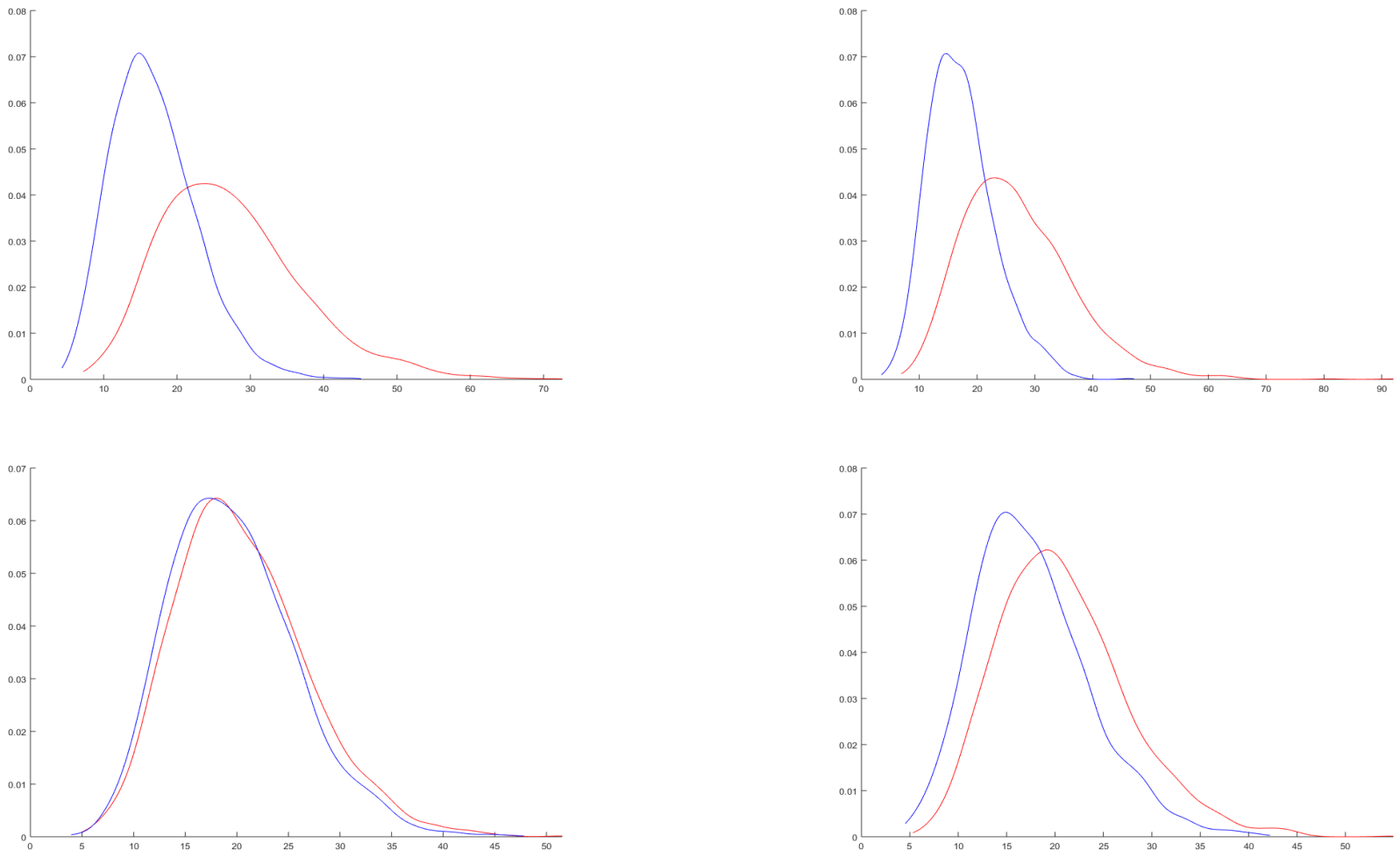
Specification testing failures



Density functions of the HJ statistic (red solid curve), weighted sum of χ^2's approximation (blue solid curves), asymptotic distribution based on Theory 12 (dot-dashed curve). Top left-hand side panel: $T = 100$, three strong factors f_t; top right-hand side panel: $T = 100$, three weak factors g_t; bottom left-hand side panel: $T = 1000$, three strong factors f_t; bottom right-hand side panel: $T = 1000$, three weak factors g_t.

Theorem (asymptotic distribution of the HJ statistic)

Under some mild assumptions, let T increase to infinity with fixed N then the behavior of the HJ statistic is characterized by:

$$T\widehat{\delta}_g^2 \to_d \widetilde{\psi}'_{B_g} M_{Q_r^{-\frac{1}{2}}\left(\eta_{B_g} + (0 \vdots \psi_{\beta_{g,2}})\right)} \widetilde{\psi}_{B_g}$$

with $\widetilde{\psi}_{B_g} = \psi_{B_g} Q_G \theta_G$.

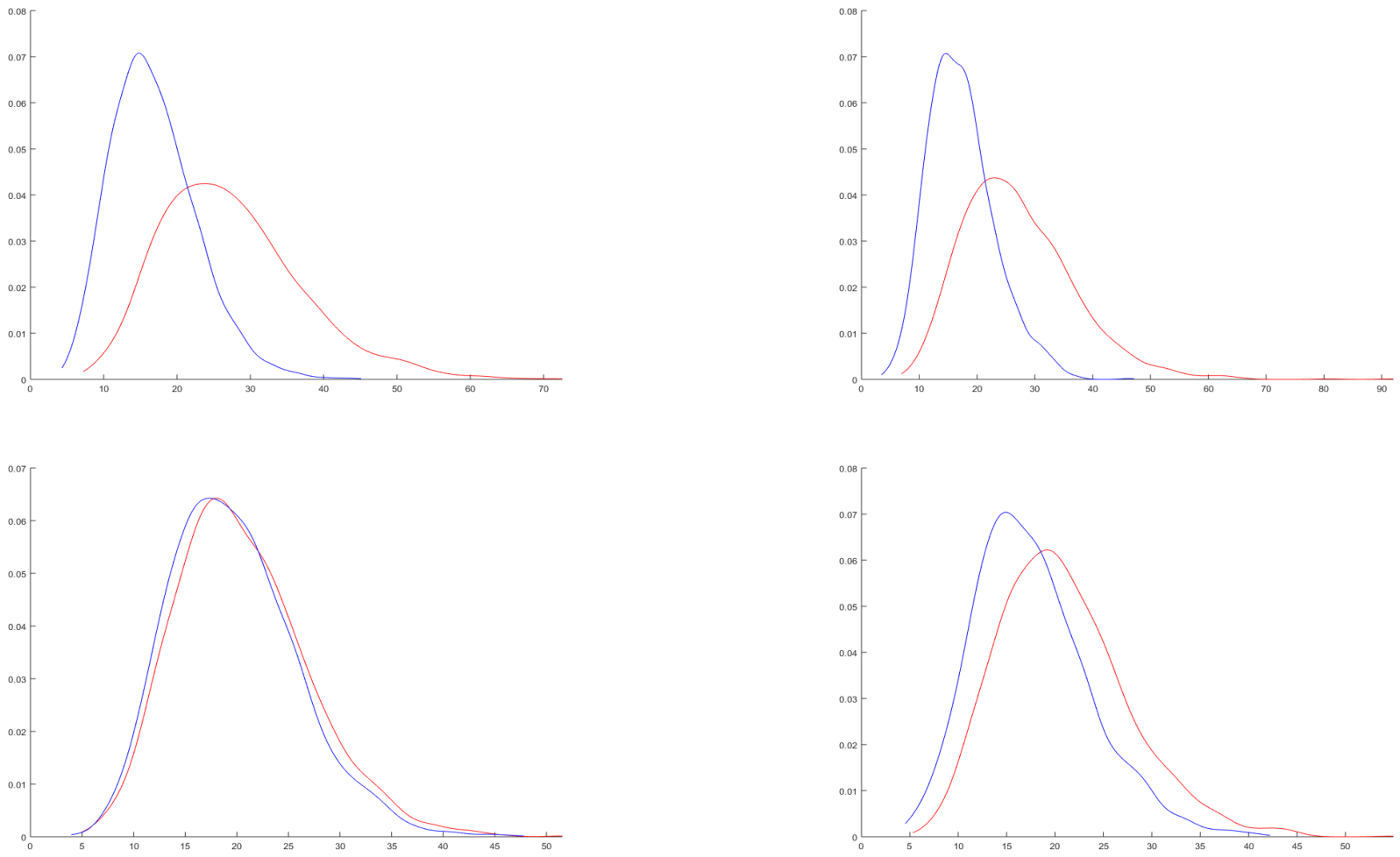
Theorem (size distortion of the HJ specification testing procedure)

Under some mild assumptions, in the presence of weak (proxy) factors

$$\limsup_N \limsup_T \ \mathbb{P}_0\left(T\widehat{\delta}_g \geq \widehat{c}_{1-\alpha}\right) = 1$$

where $\widehat{c}_{1-\alpha}$ is the conventional critical value.

Specification testing failures



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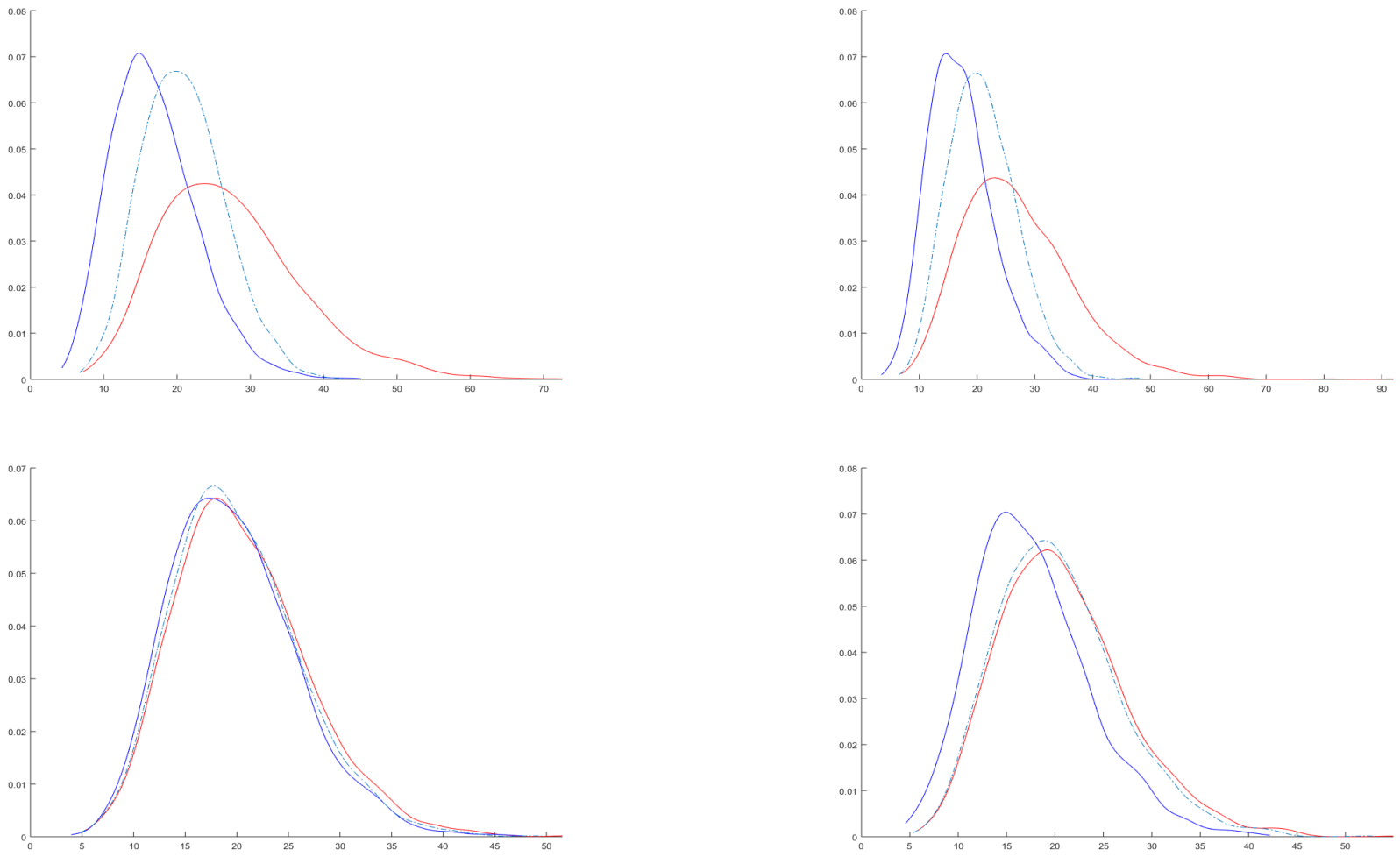

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Specification test failures

β	$N = 5$	$N = 10$	$N = 15$	$N = 31$
T=100, $d_g = 1.9$	0.5032	0.8824	0.9711	0.9992
T=10000, $d_g = 1.9$	0.1210	0.1486	0.2298	0.2238
T=100, $d_g = 0.9$	0.7132	0.9330	0.9814	1
T=10000, $d_g = 0.9$	0.5174	0.8906	0.8834	0.9978

Table: Rejection frequency table under the null (at 0.05 significance level) via the $T\widehat{\delta}_g$ with critical values drawn from weighted sum of χ^2's

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HJS statistic and critical values

$$\widehat{\delta}^* = \inf_{\theta \in CS_{r,\alpha_1}} \delta(\theta) \tag{2}$$

with CS_r is the robust confidence set constructed by inverting the AR test statistic

$$CS_{r,\alpha_1} = \left\{\theta \in \Theta : AR(\theta) \leq c_{1-\alpha_1}\right\}; \quad AR(\theta) = Te_T(\theta)'\widehat{\Sigma}^{-1}_{e_T(\theta)} e_T(\theta)$$

and the critical value of $T\widehat{\delta}^*$ is constructed as

$$c^*_{1-\alpha} = \sup_{\theta \in CS_{r,\alpha_1}} c^*_{1-\alpha_2}(\theta) \tag{3}$$

with $c^*_{1-\alpha}(\theta)$ the critical values drawn from weighted sum of N $\chi^2(1)$ random variables with weights being the non-zero eigenvalues of $(\widehat{\Sigma}_e(\theta))^{\frac{1}{2}}\,\widehat{Q}_r^{-1}(\widehat{\Sigma}_e(\theta))^{\frac{1}{2}\,\prime}$ and α_1, α_2 is chosen such that $(1-\alpha_1)(1-\alpha_2) = 1-\alpha$.

Alternative specification testing procedure: HJS statistic

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Lemma (confidence set coverage probability)

Suppose some regular assumptions hold, then even in the presence of weak identification

$$\liminf_T \mathbb{P}\left(\theta_G \in CS_{r,\alpha}\right) \geq 1-\alpha$$

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Theorem (the HJS specification test is size correct)

Suppose some regular assumptions hold, then even in the presence of weak identification

$$\limsup_T \mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) \leq \alpha$$

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Suppose some regular assumptions hold, then even in the presence of weak identification

$$\limsup_T \mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) \leq \alpha$$

Sketch of proof: $\mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) \leq \mathbb{P}\left(T\widehat{\delta}_g(\theta_G) \geq c^*_{1-\alpha}\right) \leq \mathbb{P}\left(T\widehat{\delta}_g(\theta_G) \geq c_{1-\alpha}(\theta_G)\right)$

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Example (sup does not work)

Suppose some regular assumptions hold, $K = K_{g,2} \geq 2$, then

$$\limsup_T \mathbb{P}\left(\left(\sup_{\theta \in CS_{r,\alpha_1}} \delta_{g,T}(\theta)\right) \geq c^*_{1-\alpha}\right) = 1$$

Power comparison: HJS vs J

$\mathcal{J}$ specification test

$$\mathcal{J} = \inf_{\theta} AR(\theta)$$

Power comparison: HJS vs J

$\mathcal{J}$ specification test

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Theorem: case (i) (*Gospodinov et al. 2017* unidentified model under misspecification)

Suppose some regular conditions hold. Let $H = (\iota_N, q_G)$. Suppose H has a column rank $K + 1 - k$ for an integer $k \geq 1$, then we have

$$\mathcal{J} \to_d w_{k,i}$$

where $w_{k,i}$ is the smallest eigenvalue of $\mathcal{W}_k \sim \mathcal{W}_k\left(N - K - 1 + k, I_k\right)$ and $\mathcal{W}_k\left(N - K - 1 + k, I_k\right)$ denotes the Wishart distribution with $N - K - 1 + k$ degrees of freedom and a scaling matrix I_k. Furthermore,

$$\limsup_{T} \mathbb{P}\left(\mathcal{J} \geq c_{\chi^2_{N-K},1-\alpha}\right) \leq \alpha,$$
$$\liminf_{T} \mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) = 1,$$

with $c_{\chi^2_{N-K},1-\alpha}$ the $1 - \alpha$ quantile of χ^2_{N-K}.

Power comparison: HJS vs J

$\mathcal{J}$ specification test

$$\mathcal{J} = \inf_{\theta} AR(\theta)$$

Theorem: case (ii) (*weakly identified model under misspecification*)

Suppose some regular conditions hold. Let $H = \left(\iota_N, q_G Q_G^{-1}\right)$. Suppose Suppose $(HQ_x)' * (HQ_x)$ with $Q_x = diag(I, \sqrt{T} I_{K_{g,2}})$ converges to a positive definite matrix, then we have

$$\mathcal{J} \to_d w_{k,ii}$$

where $w_{k,ii}$ is the smallest eigenvalue of $\mathcal{W}_k \sim \mathcal{W}_k\left(\mu, N-K-1+k, I_k\right)$ and $\mathcal{W}_k\left(\mu, N-K-1+k, I_k\right)$ denotes the non-central Wishart distribution with $N-K-1+k$ degrees of freedom and a scaling matrix I_k, a location parameter μ (μ is specified in the proof). Furthermore,

$$\liminf_T \mathbb{P}\left(\mathcal{J} \geq c_{\chi^2_{N-K},1-\alpha}\right) < 1,$$

$$\liminf_T \mathbb{P}\left(T\widehat{\delta}_g^* \geq c^*_{1-\alpha}\right) = 1,$$

with $c_{\chi^2_{N-K},1-\alpha}$ the $1-\alpha$ quantile of χ^2_{N-K}.

$\widehat{\theta}_G$ revisit

$$\iota_N = q_{G,T}\theta_G - \epsilon_{q_G}\theta_G,$$

with $\epsilon_{q_G} = (q_{G,T} - \mathbb{E}q_{G,T})$ being correlated with the $q_{G,T}$.

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by the following two steps: (2.1) determine the number of factors, K_{vz}, in $\widehat{u}_{g,t} = r_t - \widehat{c} - \widehat{\beta}_g \bar{g}_t$ by

$$\widehat{K}_{vz} = \arg \min_{0 \le j \le K_{vz,\max}} \left(N^{-1} T^{-1} \lambda_j \left(\widehat{u}_g \widehat{u}_g' \right) + j\phi(N, T) \right) - 1$$

where $\widehat{u}_g$ is $T \times N$ matrix stacked with the OLS residuals $\widehat{u}_{g,t}$, $K_{vz,\max}$ is an arbitrary upper bound for K_{vz} and $\phi(N, T)$ is a penalty function with the properties
$\phi(N, T) \to 0, \phi(N, T)/(N^{-\frac{1}{2}} + T^{-\frac{1}{2}}) \to \infty$; (2.2): estimate the $T \times N$ common component matrix $cc = xb'$ stacked with the common components cc_t:

$$\widetilde{cc} = \widehat{x}\widehat{b}, \quad \left(\widehat{x}, \widehat{b}\right) = \arg \min_{b_i, x_t \ s.t. \ \sum_{t=1}^{T} x_t x_t' / T = I_{\widehat{K}_{vz}}} \sum_{i,t} \left(\widehat{u}_{g,it} - b_i' x_t \right)^2$$

The estimator $\widehat{x}$ is equal to the $\sqrt{T}$ times eigenvector associated with the $\widehat{K}_{vz}$ largest eigenvalues of the matrix $\widehat{u}_g \widehat{u}_g'$, and $\widehat{b} = \widehat{x}' \widehat{u}_g / T$ corresponds to the OLS estimator regressing $\widehat{u}_g$ over $\widehat{x}$.

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Step (3): Remove the missing factor structure from regressors

$$\widetilde{q}_{G,T}^{(i)} = q_{G,T}^{(i)} - \widetilde{cc}_{G,T}^{(i)\prime}, i = 1, 2$$

where

$$q_{G,T}^{(1)} = \frac{1}{\lfloor \frac{T}{2} \rfloor} \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} r_t G_t'; \; q_{G,T}^{(1)} = \frac{1}{T - \lfloor \frac{T}{2} \rfloor} \sum_{t=\lfloor \frac{T}{2} \rfloor + 1}^{T} r_t G_t'$$

$$\widetilde{cc}_{G,T}^{(2)} = \frac{1}{\lfloor \frac{T}{2} \rfloor} \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} \widetilde{cc}_t G_t'; \; \widetilde{cc}_{G,T}^{(2)} = \frac{1}{T - \lfloor \frac{T}{2} \rfloor} \sum_{t=\lfloor \frac{T}{2} \rfloor + 1}^{T} \widetilde{cc}_t G_t'.$$

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Step (4): Our proposed four-pass estimator is then derived via IV regression where we use instrument $\widetilde{q}_{G,T}^{(1)}$ for $\widetilde{q}_{G,T}^{(2)}$ and vise versa:

$$\widetilde{\theta}_G = \sum_{i=1}^{2} \widetilde{\theta}_G^{(i)}/2,$$

with $\widetilde{\theta}_G^{(1)} = \left(\widetilde{q}_{G,T}^{(1)\prime} P_{\widetilde{q}_{G,T}^{(2)}} \widetilde{q}_{G,T}^{(1)} \right)^{-1} \widetilde{q}_{G,T}^{(1)\prime} P_{\widetilde{q}_{G,T}^{(2)}} \iota_N$ and $\widetilde{\theta}_G^{(2)} = \left(\widetilde{q}_{G,T}^{(2)\prime} P_{\widetilde{q}_{G,T}^{(1)}} \widetilde{q}_{G,T}^{(2)} \right)^{-1} \widetilde{q}_{G,T}^{(2)\prime} P_{\widetilde{q}_{G,T}^{(1)}} \iota_N$.

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Theorem (consistency)

Under regular assumptions and $N/T \to c \geq 0$

$$\sqrt{NT} Q_x^{-1} V_g \left(\widetilde{\theta}_g - \theta_g\right) \to N(0, \Sigma_{\theta_g})$$

with $Q_x = \text{diablk}\left(1, I_{K_{g,s}}, \sqrt{T} I_{K_{g,w}}\right)$. When $c = 0$ we have one estimator for Σ_{θ_g}.

Alternative specification testing procedure: HJN statistic

The HJN statistic

- Estimate θ_G from a set of N base portfolios r_t of assert returns using our proposed four-pass estimator $\widetilde{\theta}_G$

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$$\widetilde{\delta}_g^2 = \left(\iota_n - \widetilde{q}_G \widetilde{\theta}_G\right)' \widehat{Q}_R^{-1} \left(\iota_n - \widetilde{q}_G \widetilde{\theta}_G\right),$$

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Asymptotic characterization

$$T\widetilde{\delta}_g^2 = T\left(\iota_n - \widetilde{q}_G\widetilde{\theta}_G\right)' \widehat{Q}_R^{-1} \left(\iota_n - \widetilde{q}_G\widetilde{\theta}_G\right)$$

$$\sqrt{T}\left(\iota_n - \widetilde{q}_G\widetilde{\theta}_G\right) = \underbrace{\sqrt{T}\left(\iota_n - \widetilde{q}_G\theta_G\right)}_{O_p(1)} - \underbrace{\sqrt{T}\left(\widetilde{q}_G\left(\widetilde{\theta}_G - \theta_G\right)\right)}_{O_p(\frac{1}{N^{1/2}})}$$

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with $\widehat{Q}_R = \frac{1}{T}\sum_{t=1}^{T} R_t R_t'$.

Theorem (HJN specification test)

Under regular assumptions,

$$T\widetilde{\delta}_g^2 \sim_d \sum_{i=1}^{n} p_i x_i$$

with x_i being independently $\chi^2(1)$ distributed random variables and p_i being the positive eigenvalues of the matrix $S^{\frac{1}{2}}(\widehat{Q}_R)^{-1}S^{\frac{1}{2}'}$ with
$S = \frac{1}{T}\sum_{t=1}^{T} e_{g,t,R}(\widetilde{\theta}_G)e_{g,t,R}(\widetilde{\theta}_G)'$, $e_{g,t,R}(\theta_g) = \iota_N - R_t G_t'\theta_g$.

Size curves

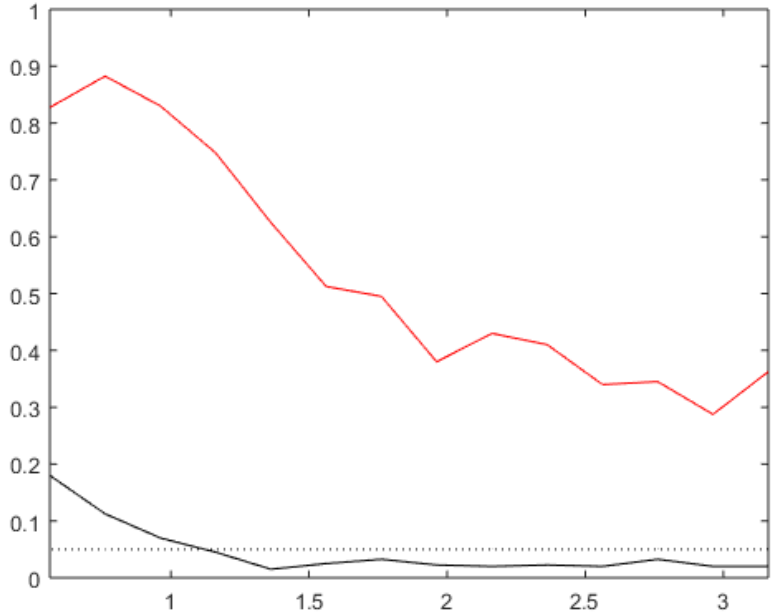

Figure: Rejection rates against the strength of the proxy factor α under the null hypothesis. Dotted curve: 5% line. Red solid curve: the HJ specification test. Black solid curves: the HJN specification test.

Alternative specification testing procedure: HJN statistic

Simulation exercise for $\widetilde{\theta}_G$

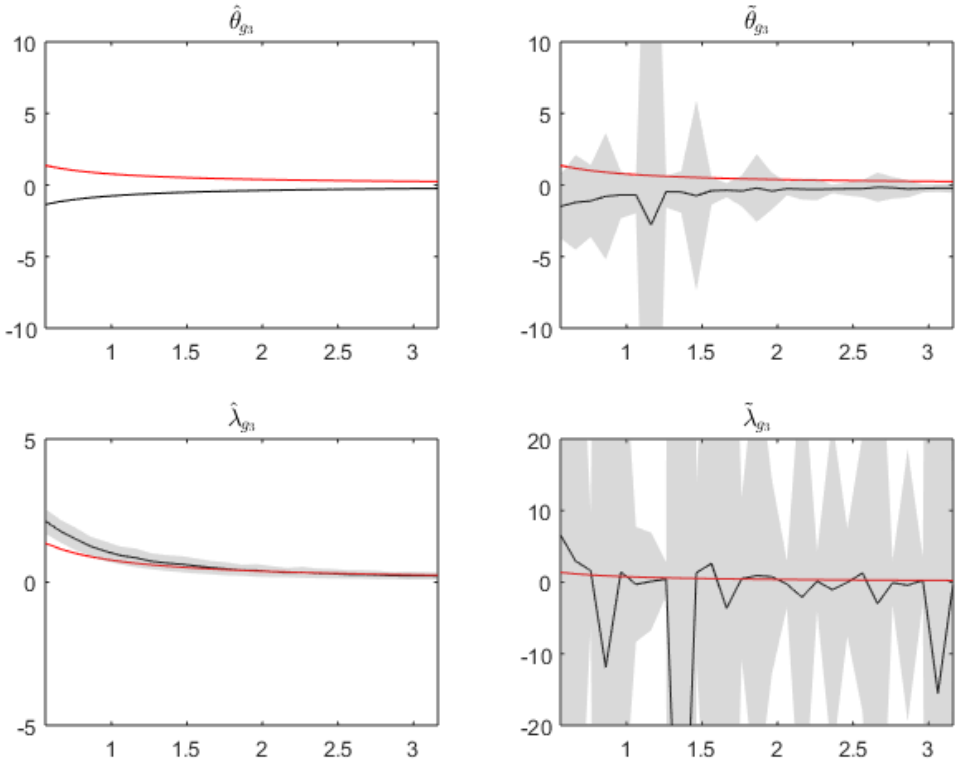

Figure: Sample mean (black solid curve) of the corresponding estimates with respect to the proxy factor $g_{3,t}$ against the strength of the proxy factor α, and the shadowed area covers one standard deviation from the sample mean. Red solid curves are the true values of the corresponding parameters. Left panel: conventional approach; right panel: our proposed approach in section 3.

literature

Linear factor models have gained tremendous popularity in the empirical asset pricing literature, Harvey et al. (2016) list hundreds of papers and factors that attempt to explain the cross-section of expected returns.

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Conclusion

Empirical application

	Market	SMB	HML	MOM	HJ(p-val)	HJN(p-val)	HJS(rejected)
(1)					0.000	0.000	Yes
$\widehat{\theta}_G$	0.0182	-0.0313	-0.0946	–			
$\widetilde{\theta}_G$	0.2467	-0.1885	0.1258	–			
$\widehat{\lambda}_g$	0.0602	0.5279	0.3742	–			
$\widetilde{\lambda}_g$	-3.2179	0.6038	0.0347	–			
(2)					**0.001**	**0.514**	No
$\widehat{\theta}_G$	0.0248	-0.0239	-0.0592	0.1952			
$\widetilde{\theta}_G$	0.1812	0.0404	0.2959	0.8543			
$\widehat{\lambda}_g$	0.3387	0.4694	0.3730	3.1841			
$\widetilde{\lambda}_g$	-3.3109	0.3053	0.0226	-10.7839			

Table: Estimates with Fama-French factors and the momentum factor and tests of specification using monthly returns on 100 portfolios sorted by size and book-to-market from 1963 to 1998

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Conclusions

- We document failures of conventional approaches when using the HJ distance statistic
- We provide novel approaches for carrying out identification robust specification test
- We provide a novel risk premia estimator.